

KAVIN T.S.

Geology Postgraduate | Geosciences & Computational Techniques

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PROFESSIONAL SUMMARY

Geology postgraduate with focused research interest in geomorphology, basin analysis, and landscape evolution. Experienced in integrating field observations with remote sensing, GIS, and Python-based computational modelling. Hands-on exposure to tectono-geomorphic analysis, DEM-based morphometrics, and seismic/geo-data workflows. Actively seeking PhD training under structured research supervision in Earth surface processes and source-to-sink systems.

EDUCATION

Qualification	Institution	Grade	Duration
M.Sc., Geology	Periyar University, Salem	7.9 CGPA	2023 – 2025
B.Sc., Geology	Government Arts College (Autonomous), Coimbatore	7.3 CGPA	2020 – 2023
HSC	Indian Matriculation Hr Sec School, Dharmapuri	71%	2020
SSLC	Indian Matriculation Hr Sec School, Dharmapuri	82%	2018

PUBLICATION

Kavin, T. S., et al. (2025). *Geochemical signatures and petrogenesis of intraplate dolerite dykes in the Salem block, Southern Granulite Terrane, India. Acta Geochimica.* <https://doi.org/10.1007/s11631-025-00830-6>

PROJECTS

M.Sc. Dissertation Periyar University	2024–2025
Geomorphic and Tectonic Influences on the Longitudinal and Cross Profiles of the Northern Thamirabarani River Basin, Southern India.	
Internship NLC, Tamil Nadu	2023
A Case Study of Lignite Exploration and Hydrological Conditions of Neyveli Basin, Tamil Nadu.	
B.Sc. Project Government Arts College, Coimbatore	2022–2023
Petrology of Punjai Puliampatti, Coimbatore District, Tamil Nadu.	

RESEARCH INTERESTS

- Sedimentology
- Geomorphology & Landscape Evolution Modelling
- Remote Sensing
- Basin Analysis & Source-to-Sink Systems
- Tectono-Geomorphic Analysis
- Carbon Sequestration

SKILLS & EXPERTISE

- **Python Applications in Geosciences**
 - Seismology & Seismic Ambient Noise Characterisation
 - Gravity & Magnetic Data Inversion
 - Electrical & Electromagnetic Data Modelling
 - Landscape Evolution Modelling (LEMPy / TopoToolbox)
 - Machine Learning with Scikit-Learn and TensorFlow
- **Remote Sensing & GIS**
 - Satellite Image Interpretation & Classification
 - Large-scale Image Analysis on Google Earth Engine (GEE)
 - Digital Elevation Model (DEM) Analysis & Morphometrics
 - LULC Mapping, Environmental Impact Assessment using NDVI, NDWI, LST, BSI
- **Field & Laboratory Skills**
 - Geological Field Mapping & Surveying
 - Fossil Collection & Identification
 - Thin Section Preparation, Optical Mineralogy & Petrographic Analysis
 - Geochemical Sampling & Analysis
- **Programming Languages**
 - Python, JavaScript, R, SQL
- **Software & Tools**
 - ArcGIS Pro, QGIS, Jupyter Notebook, Google Colab, MATLAB, Adobe Illustrator, MS Office
- **Operating Systems**
 - Windows, Linux (CentOS), macOS

PAPER PRESENTATIONS

- Tectono-Geomorphic Evolution of river profiles in Thamirabarani River Basin - Aspiring Geologists' Extravaganza '25, Anna University.
- Geomorphic and Tectonic Influences on the Longitudinal and Cross Profiles of Thamirabarani River Basin, Southern India - Central University of Tamil Nadu.
- Enhancing Coastal Ecosystems Through LULC and Sediment Management for Climate Resilience and Biodiversity - VIKSIT Bharat @2047 (ISSC-E3).
- Utilizing Mineral Carbonation for CO₂ Sequestration to Combat Climate Change - VIKSIT Bharat @2047 (ISSC-E3).
- A Bibliometric Analysis of CCS Techniques, Applications and Regional Trends - International Conference on Subsurface Technology (ICSST-2024), September 2024.

FIELD EXPERIENCE

Undergraduate

Participated in a six-day geological expedition across Tamil Nadu covering KVK Mine (Dalmia Private Limited), Terani Clay, Neyveli Lignite Mine (NLC), Sri Ponguru Magnesite Mines, and India Cements. Explored geological sites including Sangagiri Fort, Bauxite deposits in Shevroy Hills, and the Sittampoondi Anorthosite Complex.

Postgraduate

Advanced explorations including fossil collection, visits to TANCEM Limestone Mines and the Geological Museum in Ariyalur; study of the Niniyur Formation and coastal features of Rameswaram island. Independently conducted basin analyses of the Cauvery, Thamirabarani, and Vaigai basins. Collaborated with the Birbal Sahni Institute of Palaeosciences on East Coast shoreline dynamics and ecosystem changes.

TRAINING & ACADEMIC EVENTS

- Python in Geoscience - CSIR-NGRI, Hyderabad (CSIR Integrated Skill Initiative Program), January 2025
- Aspiring Geologists' Extravaganza '25 - Society of Geologists, Anna University, March 2025
- Microplastic Processing and Risk Assessment - Workshop, National College, Trichy, March 2025
- GeoVista 2025: Geographical Visualization - National Conference, Central University of Tamil Nadu, March 2025
- Hyperspectral Remote Sensing Workshop - Centre for Geoinformatics and Planetary Science, Periyar University
- International Conference on Climate Change and Environmental Sustainability (CCES-2024), January 2024
- WINGS (World Innovative Neoteric Geological Science) Conference - Periyar University, March 2024
- Geological Field Mapping Workshop - Periyar University, September 2022
- G20 and Geo Expo 2023: Sustainable Future Themes, October 2023
- Soft Skills Development: Employability Training - IBM Skills Build & Govt. of Tamil Nadu, 2022-23

NATIONAL CADET CORPS (NCC) & HONOURS

- DG NCC Commendation Card - Awarded on Indian Army Day, 15 January 2023, by the Ministry of Defence.
- Silver Medal - 50m .22 Open Sight Rifle Shooting, Inter-Group Shooting Competition, TN, P&AN DTE, NCC GP HQ Puducherry.
- Represented at Inter-DTE Sports Shooting Championship, Indore, MP (23 Sept – 2 Oct 2021).
- NCC 'C' Certificate - Grade A | NCC 'B' Certificate - Grade A
- Annual Training Camp (ATC-1) - Karunya Institute of Technology, Coimbatore, December 2021
- Combined Annual Training Camp (CATC) - Dr. NGP Arts and Science College, Coimbatore, January 2023
- Combined Annual Training Camp - NCC GP HQ Puducherry, September 2021